

Curriculum Vitae

Riccardo Colletti



Personal Information

<i>Surname</i>	Colletti
<i>Name</i>	Riccardo
<i>Address</i>	Via Marco Polo 37, 10129 Torino, Italy
<i>Telephone</i>	+39 345 780 7289 (Italy) +1 510 570 9580 (US)
<i>E-mail</i>	riccardo_colletti@berkeley.edu ing.colletti.riccardo@gmail.com
<i>LinkedIn</i>	linkedin.com/in/riccardo-colletti
<i>Website</i>	collettiriccardo.com
<i>Nationality</i>	Italian
<i>Date of birth</i>	17 December 1998

Education and Training

My study path began at *Liceo Scientifico Internazionale* in Italy (100/100), including one academic year completed in Ireland at Coláiste Chríost Rí. I then obtained a BSc in *Electronic and Communications Engineering* at Politecnico di Torino (110/110 cum laude), a Master in *Marketing and Communication* at Bocconi University (110/110 cum laude, top of cohort), and an MEng in *Electrical Engineering and Computer Sciences* at UC Berkeley (GPA 4.0/4.0). Each programme was completed on a merit-based scholarship.

Drawn by the research culture of Berkeley and the Bay Area, in the past year I have channelled my work into **machine learning and reinforcement learning**, co-leading **paper submissions to top-tier venues** on reinforcement-learning theory, multi-agent architecture and robustness to distribution shift.



University of California, Berkeley

MEng in Electrical Engineering and Computer Sciences

<i>Date (from – to)</i>	August 2025 – May 2026
<i>Name and organisation</i>	University of California, Berkeley College of Engineering, EECS Department Berkeley, CA 94720, USA
<i>Programme duration</i>	One-year Master of Engineering programme (full-time).
<i>Principal subjects covered</i>	Coursework focused on statistical learning theory (CS C281A, Recht), optimisation (EECS 227AT, Sojoudi), machine learning (CS 289A, Norouzi & Gonzalez), deep reinforcement learning (CS 285, Levine), and robotics (EECS C206A, Tennant). <i>Master of Engineering Capstone</i> (ENGIN 296MA/MB, advisor Prof. Paul Grigas, IEOR): team applied-research project on robust sequential decision-making under non-stationary conditions, combining predictive machine learning with constrained optimisation. <i>Fung Institute leadership track</i> : technology strategy, R&D management, communications, teaming, organizational behaviour, marketing & product management.
<i>Title of qualification</i>	Master of Engineering (MEng) in Electrical Engineering and Computer Sciences
<i>Final mark</i>	Current GPA: 4.0/4.0 (scale 0–4.0)
<i>Scholarship</i>	Fung Excellence Scholarship – merit-based award granted by the Berkeley Engineering Fung Institute to top MEng candidates.

<i>Date (from – to)</i>	September 2023 – May 2025
<i>Name and organisation</i>	<i>Bocconi University</i> Department of Marketing Via Sarfatti 25, 20136 Milano, Italy
<i>Programme duration</i>	One-and-a-half-year full-time Master Universitario programme.
<i>Principal subjects covered</i>	Quantitatively driven programme focused on marketing analytics, consumer behaviour, pricing strategy, quantitative methods and data-informed decision-making.
<i>Title of qualification</i>	Master Universitario di primo livello in Marketing and Communication (MiMeC)
<i>Final mark</i>	110/110 cum laude (scale 66–110). Ranked first in the cohort.
<i>Scholarship</i>	Merit-based scholarship awarded for outstanding academic excellence.



<i>Date (from – to)</i>	September 2017 – July 2021
<i>Name and organisation</i>	<i>Politecnico di Torino</i> Department of Electronics and Telecommunications Corso Duca degli Abruzzi 24, 10129 Torino, Italy
<i>Programme duration</i>	Three-year Bachelor of Science programme (English-taught, honours).
<i>Principal subjects covered</i>	Electronic systems, communication networks, digital transmission, algorithms and programming, signals and systems, mathematics and physics.
<i>BSc thesis</i>	<i>Digital Phase-Locked Loop (DPLL) Simulation.</i>
<i>Honors programme</i>	Percorso Intraprendenti (formerly <i>Percorso per i Giovani Talenti</i>) – selected among the top 200 students university-wide.
<i>Title of qualification</i>	Bachelor of Science (BSc) in Electronic and Communications Engineering
<i>Final mark</i>	110/110 cum laude (scale 66–110).
<i>Scholarship</i>	Merit-based scholarship awarded for outstanding academic excellence.



<i>Date (from – to)</i>	September 2012 – July 2017
<i>Name and organisation</i>	<i>Istituto Nostra Signora delle Grazie</i> V.le Don Bosco 40 14049 Nizza Monferrato (AT), Italy
<i>Programme duration</i>	Five-year Italian secondary-school programme.
<i>Principal subjects covered</i>	Mathematics, physics, chemistry, biology, Italian and Latin literature, philosophy, history, English (advanced international track).
<i>Exchange year abroad</i>	One academic year at Coláiste Chríost Rí , Capwell Road, Turners Cross, Cork (Ireland).
<i>Title of qualification</i>	Italian high-school diploma (<i>maturità scientifica</i> , Esame di Stato)
<i>Final mark</i>	100/100 (scale 60–100).
<i>Award</i>	<i>Diploma di Merito</i> , Municipality of Nizza Monferrato (Nov 2017) – for outstanding academic dedication.

Graduation Thesis

<i>Title</i>	No Free Lunch for Food Delivery: Optimizing Courier Dispatch Under Shifting Operating Conditions
<i>Language</i>	English
<i>Supervisor</i>	Prof. Paul Grigas , IEOR Department, UC Berkeley (pgrigas@berkeley.edu).
<i>Thesis Summary</i>	Food-delivery courier dispatch, the recurring assignment of pending orders to available couriers every few seconds, is a real-time sequential decision-making task carried out under continuously shifting operating conditions. The existing literature, spanning operations-research heuristics, value-based and policy-gradient reinforcement-learning policies, and graph-attention neural matchers, evaluates each policy under a single fixed regime against a fixed baseline. Their robustness across regimes has remained largely unexamined. The thesis introduces PRISM (<i>PRediction-Informed Smart Matching</i>), a predict-then-optimize dispatch architecture that decomposes the end-to-end assignment into two independently trained learning components, one estimating per-pair assignment costs and one anticipating upcoming demand, combined through an exact constrained-matching solver. The decomposition produces the two contributions of the work. First, the architecture is measurably more robust when operating conditions shift, outperforming both heuristic and learning-based baselines under controlled regime ablations. Second, it is modular along the objective dimension: secondary objectives can be added at inference time without retraining the underlying cost model, exposing the design trade-offs that a single end-to-end policy would otherwise hard-code.

Publications and Articles Submitted

Asterisks (*) denote equal contribution. The two most significant submissions are enclosed as attachments (Annex 1 and Annex 2).

<i>Author(s) and title</i>	Holzinger, R.* & Colletti, R.* <i>‘HydraMARE: Isolating the Parameter-Sharing Spectrum in Multi-Agent Reinforcement Learning’</i> (included as attachment to this application). The paper introduces a principled architectural framework that decomposes per-agent policy and value networks into a population-shared trunk and per-agent heads, applied symmetrically across actor and critic. A single continuous parameter governs the full spectrum between complete parameter sharing and full agent independence, recovering both extremes as limit cases. Systematic experiments on the VMAS multi-agent benchmark suite identify the optimal sharing regime and characterise its sensitivity to task structure, degree of cooperation and population size, providing the first comprehensive empirical map of the sharing–specialisation trade-off in cooperative multi-agent reinforcement learning.
<i>Language</i>	English
<i>Publication place</i>	Submitted to NeurIPS 2026 (main track, CORE A*) – currently under review.
<i>Date of publication</i>	Submitted May 2026 (under review)

<i>Author(s) and title</i>	Colletti, R.* & Holzinger, R.* <i>‘KLip-PPO: A per-sample KL perspective on PPO-Clip’</i> (klip-ppo.org, GitHub). The paper establishes that the gradient of the PPO-Clip surrogate objective is exactly equivalent to that of a Kullback-Leibler-penalised surrogate with a per-sample, state-dependent coefficient. The equivalence is derived analytically and validated empirically on standard MuJoCo continuous-control benchmarks, providing a principled reinterpretation of PPO-Clip as an adaptive trust-region method and opening a direct route to per-sample regularisation schedules with explicit theoretical guarantees.
<i>Language</i>	English
<i>Publication place</i>	Preprint (arXiv:2606.23932), planned submission to a NeurIPS 2026 Workshop (NeurIPS is CORE A*).
<i>Date of publication</i>	Preprint 22 June 2026, planned workshop submission summer 2026

<i>Author(s) and title</i>	Holzinger, R.* , Colletti, R.* & Remmerie, A. <i>'Drift Happens: An Empirical Study of Neural Architecture Robustness to Temporal Distribution Shift'</i> (drift-happens.org). A systematic empirical study of how neural architectures respond to temporal distribution shift across vision (Yearbook portraits) and text (arXiv title+abstract, Amazon Reviews) domains. The study shows that in-distribution accuracy does not predict drift robustness: architectures with strong domain-aligned inductive biases peak in-distribution but degrade fastest, while frozen pretrained encoders trade accuracy for long-horizon stability.
<i>Language</i>	English
<i>Publication place</i>	Preprint; submitted to QCDS 2026 – 1st International Workshop on Quantification and Classification under Dataset Shift, at ECML-PKDD 2026 – currently under review.
<i>Date of publication</i>	Submitted 6 June 2026 (under review)

Language Certifications



IELTS – International English Language Testing System

<i>Issued by</i>	IELTS Official
<i>Result</i>	English Language Certificate, Band 8/9
<i>Validity</i>	Issued Aug 2024 – Valid until Aug 2026

Professional Certifications



Massachusetts Institute of Technology (MIT IDSS)

<i>Programme</i>	<i>Data Science and Machine Learning: Making Data-Driven Decisions.</i>
<i>Date</i>	May 2025 – Oct 2025.



Google Analytics

<i>Programme</i>	<i>Google Analytics Certification.</i>
<i>Date</i>	Jul 2024.



Università Bocconi Professional certifications

<i>Programme</i>	<i>Text Analysis with Python.</i>
<i>Date</i>	Jul 2024.

<i>Programme</i>	<i>R for Data Analysis.</i>
<i>Date</i>	Oct 2024.

<i>Programme</i>	<i>Cybersecurity: Risks, Countermeasures, and Compliance.</i>
<i>Date</i>	Oct 2024.

<i>Programme</i>	<i>Digital Marketing Tools.</i>
<i>Date</i>	Jul 2024.

<i>Programme</i>	<i>Photoshop Elements.</i>
<i>Date</i>	Mar 2024.

Work Experience, Stages, Studies Abroad

 **Lenovo Italy S.r.l.**
Multinational technology company (PCs, smart devices, IT services)

Date (from – to)	January 2025 – July 2025
Address	Milan, Italy
Type of employment	Marketing Specialist, CRM Digital Services (employment contract)
Main activities and responsibilities	Data-driven initiatives at the intersection of customer data, product strategy and commercial execution. Customer segmentation, campaign analysis, CRM data exploration. Cross-functional collaboration with marketing, sales and external partners in a large-scale corporate environment.

 **Rialco S.R.L.**
Digital consulting firm (web, e-commerce, growth) for SMB clients

Date (from – to)	2021 – 2025
Address	Milan, Italy
Type of employment	CEO / Product Manager / Growth Manager
Main activities and responsibilities	As CEO , owned strategic direction and P&L of a digital consulting firm, coordinating an extended network of external collaborators (developers, designers, marketing partners) to deliver web, e-commerce and growth solutions to SMB clients. As Product Manager , led discovery, roadmap and end-to-end delivery of digital products. As Growth Manager , ran acquisition and conversion initiatives across SEO, paid and social.



Biesse Sistemi S.r.l.
Industrial software development (ICT, Computer Telephony Integration)

Date (from – to)	January 2020 – July 2020
Address	Piazza Garibaldi 2, 14049 Nizza Monferrato (AT), Italy
Type of employment	Software Development Intern
Main activities and responsibilities	Supported the development of a voice-controlled application for automated taxi booking, contributing to early-stage application logic and system integration. Hands-on exposure to software development and applied problem-solving.



Coláiste Chríost Rí, Cork, Ireland
Study abroad – one full academic year of secondary education

Date (from – to)	August 2015 – June 2016
Address	Capwell Road, Turners Cross, Cork, Ireland
Type of programme	Full immersion academic year in an English-speaking secondary school, integrated within the Italian <i>Liceo Scientifico Internazionale</i> curriculum.
Main activities and achievements	Followed the standard curriculum (English, mathematics, physics, chemistry, biology, geography) alongside Irish peers, and played in the school basketball team. The experience consolidated near-native English proficiency and cross-cultural communication skills.

Personal Skills and Competences

Acquired in the course of life and career but not necessarily evidenced by formal certificates and diplomas.

<i>Mother tongue</i>	Italian
<i>Other languages</i>	English – excellent in reading, writing and speaking. Certified IELTS Band 8/9 (Aug 2024). French – elementary.

Engineering & research

<i>Programming languages</i>	Python, C/C++, MATLAB.
<i>Code & development tooling</i>	Git, conda/pip/uv, pytest, mypy, ruff.
<i>ML infrastructure</i>	Weights & Biases (experiment tracking), Modal (cloud compute), Streamlit (demos).
<i>Scientific & ML libraries</i>	PyTorch, Hugging Face transformers, scikit-learn, LightGBM, XGBoost, NumPy, pandas, SciPy, SymPy.
<i>Optimisation</i>	CVXPY, convex programming, constrained matching solvers.
<i>Reinforcement-learning algorithms (implemented from scratch)</i>	PPO (clip, KL), GRPO, REINFORCE, DQN, SAC, CQL, fitted-Q-iteration, behavioural cloning, DAgger.
<i>Reinforcement-learning environments & frameworks</i>	Gymnasium, stable-baselines3, MuJoCo, VMAS, OG-Bench, highway-env.
<i>Robotics frameworks</i>	ROS 2, OpenCV, ikpy.
<i>Document preparation</i>	LaTeX.

Data analytics, marketing & design

<i>Data analytics & BI</i>	Microsoft Excel, R, Power BI; A/B testing; descriptive and inferential statistics.
<i>CRM & marketing analytics</i>	CRM data exploration, campaign analysis, customer-journey design, Google Analytics.
<i>Digital advertising & SEO</i>	Meta Ads, Google Ads, LinkedIn Ads, TikTok Ads; SEO; conversion-rate optimisation.
<i>Web & e-commerce</i>	WordPress, Shopify, HTML/CSS.
<i>Design</i>	Figma, Adobe Photoshop, Adobe Illustrator, Canva. Acquired through professional work at Rialco and through Bocconi professional certifications.
<i>Productivity & collaboration</i>	Microsoft Office, Google Workspace, Notion, Slack.

Technical skills acquired through university coursework and research projects (Politecnico di Torino, Bocconi, UC Berkeley) and professional experience (Rialco, Lenovo).

Social skills

<i>Cross-cultural adaptability</i>	Study and work across Italy, Ireland and the United States, in academic, corporate and sport contexts.
<i>Cross-functional coordination</i>	Coordinated small cross-functional teams of developers, designers and marketers at <i>Rialco</i> ; marketing, sales and external partners at <i>Lenovo</i> .
<i>International research collaboration</i>	Co-authored three papers with international collaborators for <i>NeurIPS / ECML-PKDD 2026</i> .
<i>Technical communication</i>	Translating technical work for non-technical audiences across mixed teams.

Organisational & leadership skills

<i>Executive leadership</i>	CEO of Rialco S.R.L. (2021–2025): strategic direction and full profit-and-loss accountability of a digital consulting firm.
<i>Pitching & business development</i>	Client and partner pitches; contract negotiation and scoping.
<i>Team building</i>	Recruited and coordinated a network of developer, designer and marketing collaborators.
<i>Research project management</i>	Led planning, experiments and writing for two <i>NeurIPS 2026</i> submissions.
<i>Public speaking</i>	Regular delivery of technical and business presentations in academic and corporate settings.

Additional Information

Capstone Showcase

Capstone Final Presentation	Berkeley Fung Institute, presented before the EECS Industry Advisory Board (1 May 2026). <i>PRISM: a modular predict-then-optimize architecture for robust real-time courier dispatch under operating-condition shifts</i> , joint applied-research project under the supervision of Prof. Paul Grigas (IEOR, UC Berkeley).
-----------------------------	---

Teaching Support

Private tutoring	Independent tutor in mathematics and physics to high-school students and to first-year undergraduates at Politecnico di Torino (2018–2021).
------------------	--

Scholarships

Merit-based scholarships	Fung Excellence Scholarship , UC Berkeley Fung Institute – merit-based award to top MEng candidates. Merit-based scholarship, Bocconi University (MiMeC). Merit-based scholarship, Politecnico di Torino.
--------------------------	--

Competitions and Awards

Italian Mathematical Olympiad	National finalist, <i>Olimpiadi di Matematica a Squadre</i> (Italian Mathematics Olympiad – Team Competition), national finals (school year 2015/2016). Recalled from the academic-year exchange in Ireland for one week to represent <i>Istituto Nostra Signora delle Grazie</i> .
-------------------------------	---

Sport

Discipline	Padel
Level	Professional athlete, sponsored by <i>Bullpadel</i> , ranked Top 200 Italy (2024).

Volunteering

OFTAL	Volunteer since 2010 (15+ years), supporting pilgrimages that bring ill and economically disadvantaged people to Lourdes. Recently appointed <i>Transport Coordinator</i> .
Fondazione Banco Alimentare Onlus	Food selection and distribution to families in need (2014–2018).
AGAPE Onlus	Food sorting and distribution for the local community (2014).

Research Outputs

- *HydraMARL* (included as attachment, Annex 1) – Holzinger, R.* & Colletti, R.* *Submitted to NeurIPS 2026* (main track, CORE A*).
- *KLip-PPO* ([klip-ppo.org](#), [GitHub](#)) – Colletti, R.* & Holzinger, R.* Preprint ([arXiv:2606.23932](#)), planned submission to a NeurIPS 2026 Workshop (NeurIPS: CORE A*).
- *Drift Happens* ([drift-happens.org](#)) – Holzinger, R.*, Colletti, R.* & Remmerie, A. *Submitted to QCDS 2026* (1st International Workshop on Quantification and Classification under Dataset Shift, at ECML-PKDD 2026).
- *Hold UR Pla7e* ([holdurpla7e.org](#), [GitHub](#)) – ball-on-plate stabilisation on a UR7e robotic arm, combining ArUco-based vision tracking with PID, MPC and a reinforcement-learning residual policy; validated in MuJoCo and on hardware.

Online Resources

Personal website	collettiriccardo.com
LinkedIn	linkedin.com/in/riccardo-colletti
GitHub	github.com/riccardocolletti-berkeley

Annexes

- **Annex 1** – *Holzinger_Colletti_HydraMARL_NeurIPS2026.pdf*. Holzinger, R.* & Colletti, R.* ‘HydraMARL: Isolating the Parameter-Sharing Spectrum in Multi-Agent Reinforcement Learning.’

- **Annex 2** – *Colletti_Holzinger_KLipPPO_NeurIPS2026.pdf*. Colletti, R.* & Holzinger, R.* ‘KLip-PPO: A per-sample KL perspective on PPO-Clip.’